

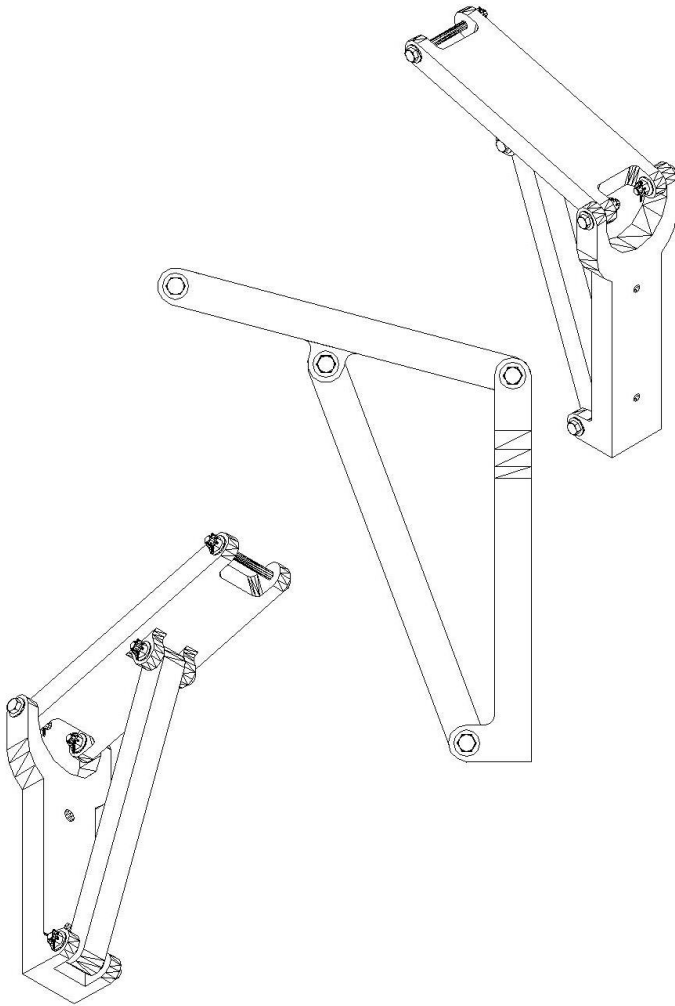
Technical Graphics

Working Drawings

- ☐ Assemblies
- ☐ Parts / Details
- ☐ Other

Isometric Views

Front View

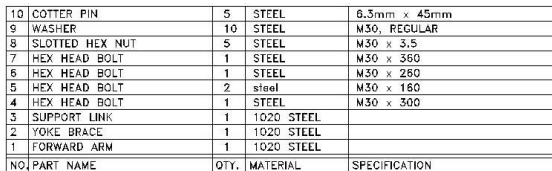


DATE:	TITLE: DESIGN DRAWINGS EXAMPLE	PAGE 1 OF 13
NÖ.:	INSTR.:	DR. BY: JcS CH. BY:
SEC.:		REV. NO.: 051994

4/5/2009 - jcs

Working Drawings

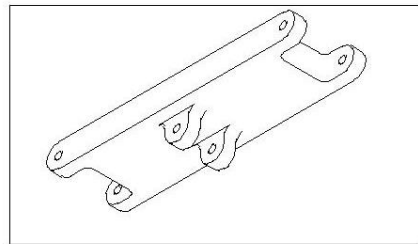
- ☐ Ballons
- ☐ Parts List
- ☐ Bill of Materials



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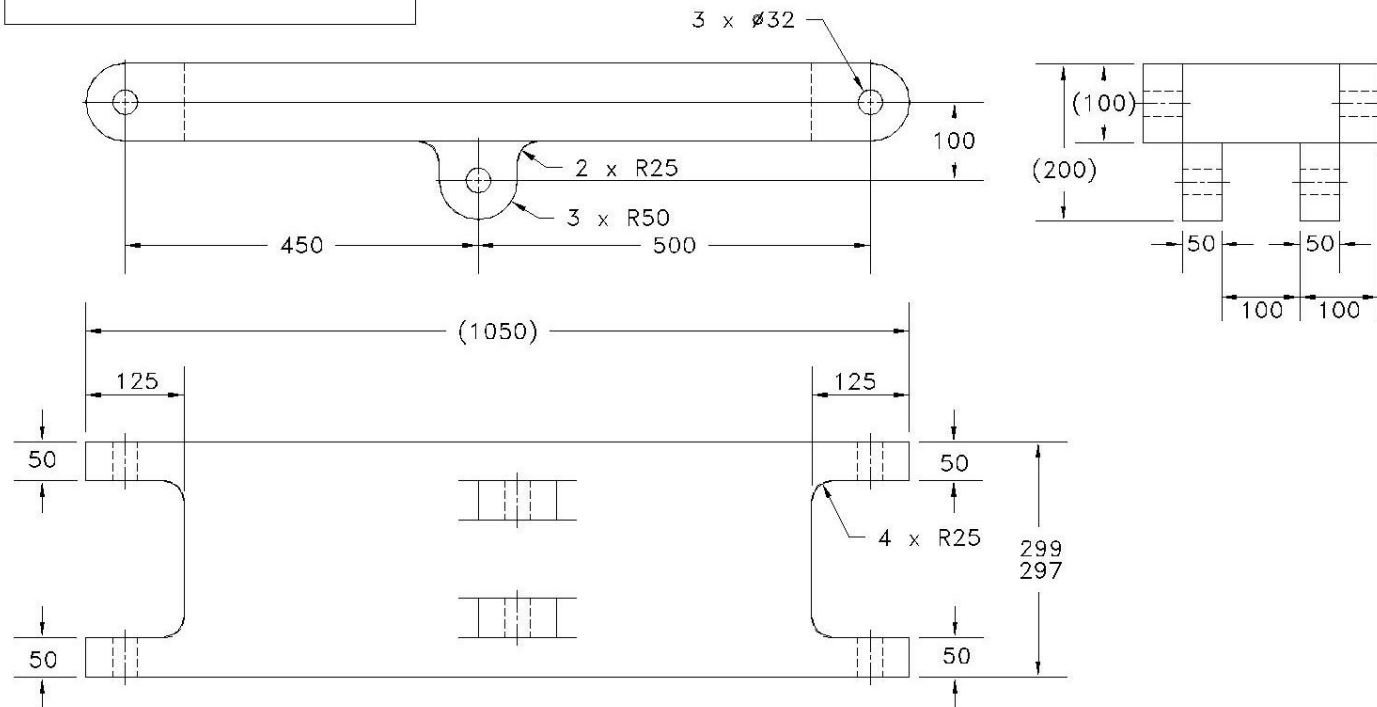
Part 1: Forward Arm

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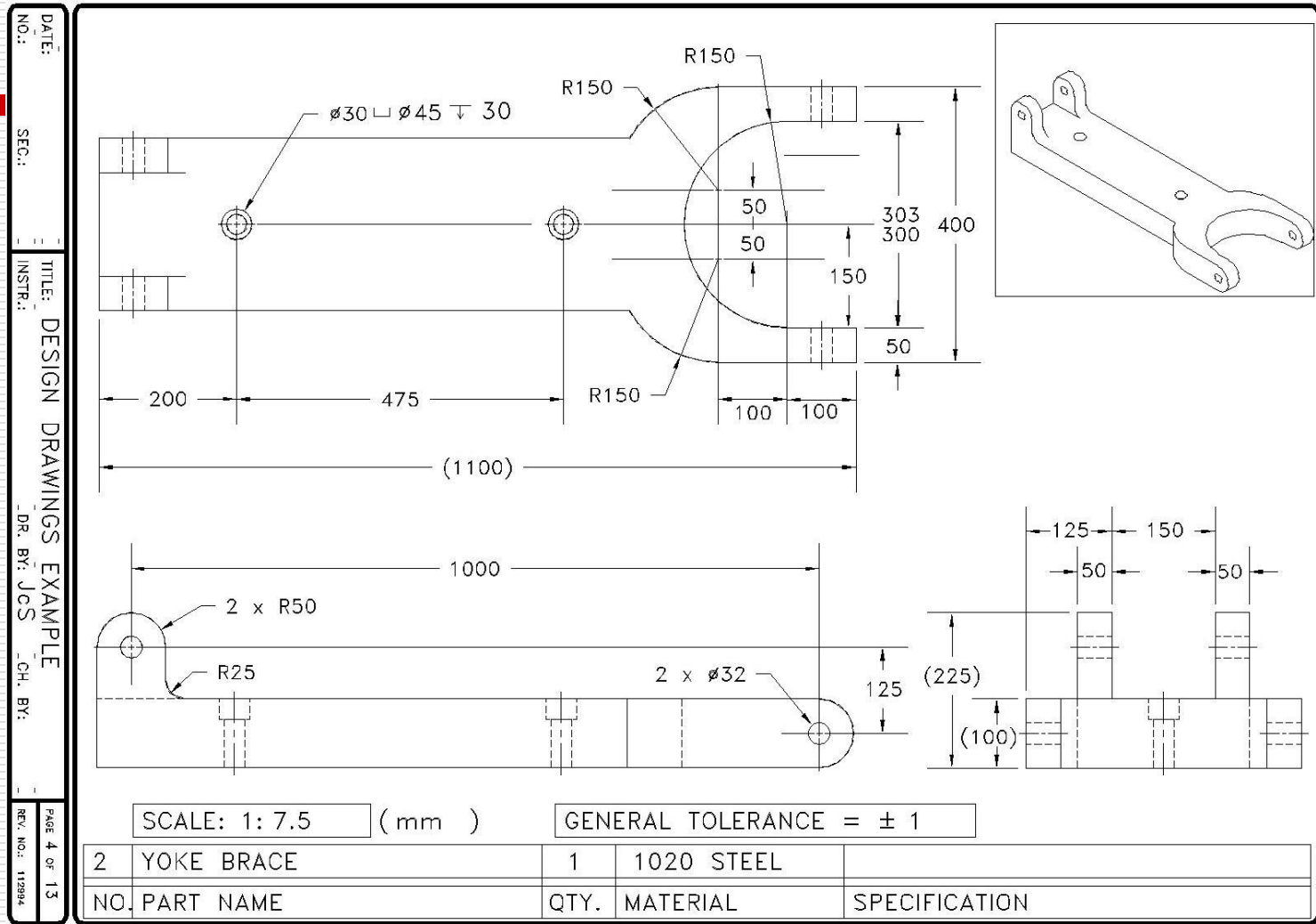
SCALE: 1: 7.5 (mm)

GENERAL TOLERANCE = ± 1



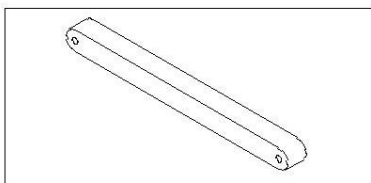
1	FORWARD ARM	1	1020 STEEL	
NO.	PART NAME	QTY.	MATERIAL	SPECIFICATION

Part: 2: Yoke Brace



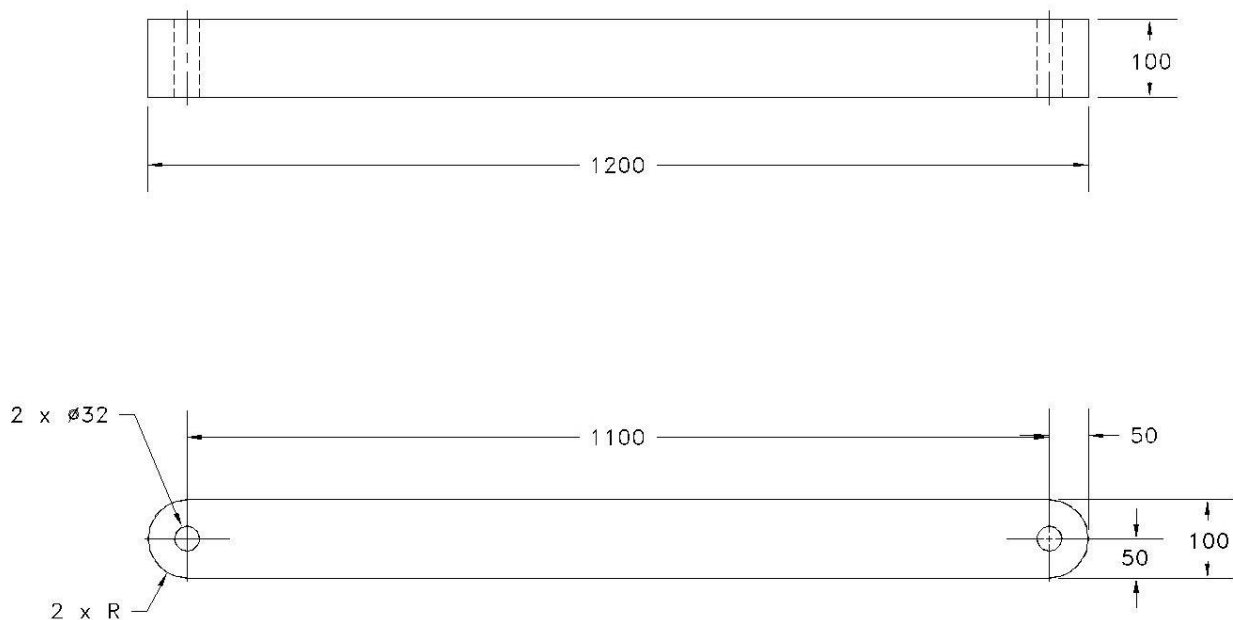
Part 3: Support Link

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SCALE: 1: 7.5 (mm)

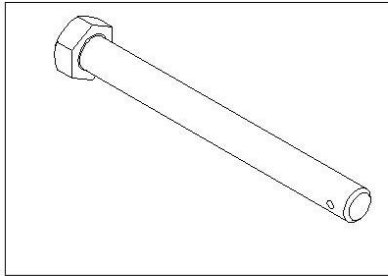
GENERAL TOLERANCE = ± 1



3	SUPPORT LINK	1	1020 STEEL	
NO.	PART NAME	QTY.	MATERIAL	SPECIFICATION

Part 4: Hex Bolt

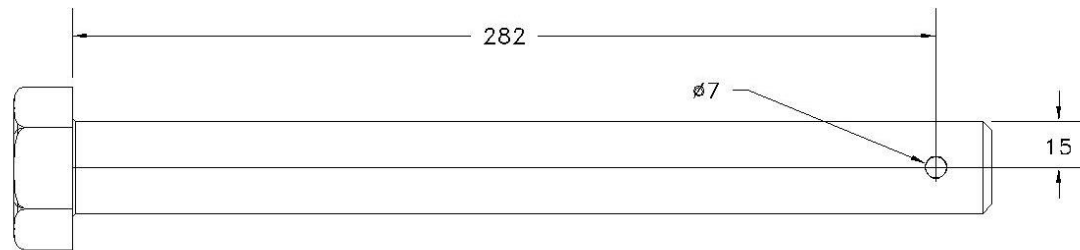
DATE: _____ NO.: _____ SEC.: _____ TITLE: DESIGN DRAWINGS EXAMPLE DR. BY: JCS CH. BY: _____ PAGE 6 OF 13 REV. NO.: 001994



SCALE: 1: 2 (mm)

GENERAL TOLERANCE = ± 1

STANDARD PART
MODIFY AS SHOWN



4	HEX HEAD BOLT	1	STEEL	M30 x 300
NO.	PART NAME	QTY.	MATERIAL	SPECIFICATION

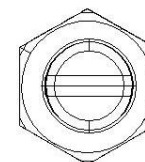
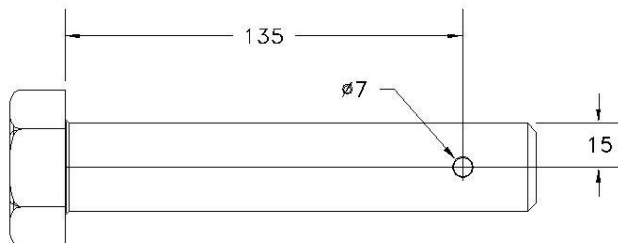
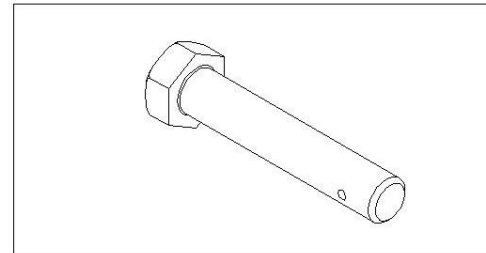
Part 5: Hex Bolt (160mm)

DATE: _____
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 TITLE: DESIGN DRAWINGS EXAMPLE
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 CH. BY: _____
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SCALE: 1: 2 (mm)

GENERAL TOLERANCE = ± 1

STANDARD PART
 MODIFY AS SHOWN



5	HEX HEAD BOLT	2	steel	M30 x 160
NO.	PART NAME	QTY.	MATERIAL	SPECIFICATION

Part 6: Hex Bolt (260mm)

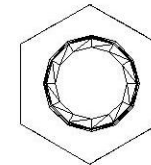
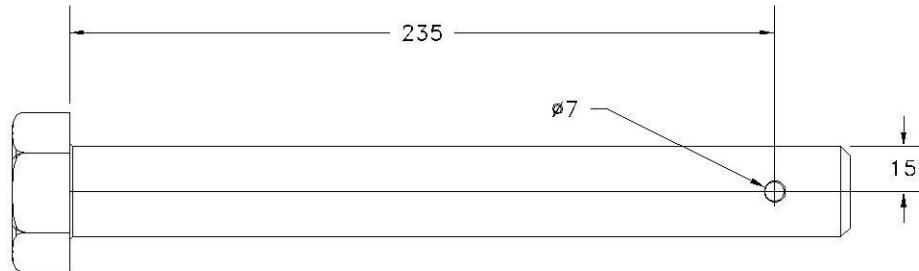
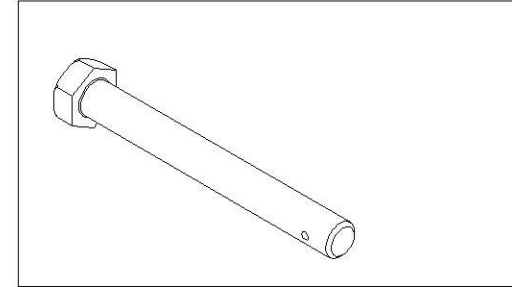
DATE: _____
 NO.: _____
 SEC.: _____
 TITLE: DESIGN DRAWINGS EXAMPLE
 INSTR.: _____
 DR. BY: JCS
 CH. BY: _____
 PAGE 8 OF 13
 REV. NO.: 051984

SCALE: 1: 2 (mm)

GENERAL TOLERANCE = ± 1

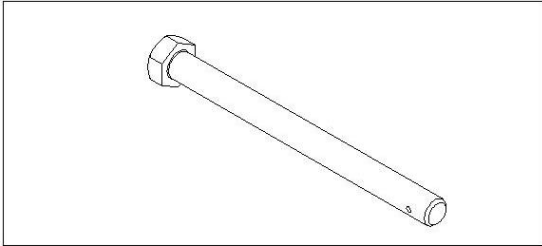
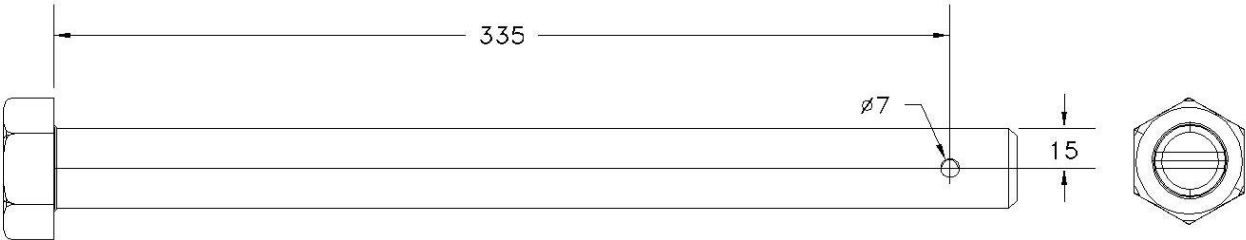
STANDARD PART

MODIFY AS SHOWN



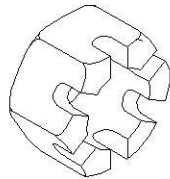
6	HEX HEAD BOLT	1	STEEL	M30 x 260
NO.	PART NAME	QTY.	MATERIAL	SPECIFICATION

Part 7: Hex Bolt (360 mm)

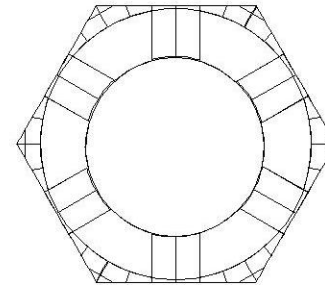
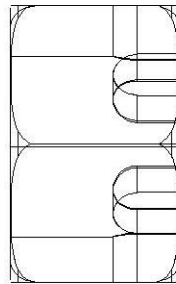
DATE: NO.:	SCALE: 1: 2.5 (mm)													
SEC.:	GENERAL TOLERANCE = ± 1													
TITLE: INSTR.:	STANDARD PART MODIFY AS SHOWN													
DR. BY: JCS CH. BY:														
REV. NO.: 051934	PAGE 9 OF 13	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 5%;">7</td> <td style="width: 40%;">HEX HEAD BOLT</td> <td style="width: 5%;">1</td> <td style="width: 20%;">STEEL</td> <td style="width: 30%;">M30 x 360</td> </tr> <tr> <td>NO.</td> <td>PART NAME</td> <td>QTY.</td> <td>MATERIAL</td> <td>SPECIFICATION</td> </tr> </table>			7	HEX HEAD BOLT	1	STEEL	M30 x 360	NO.	PART NAME	QTY.	MATERIAL	SPECIFICATION
7	HEX HEAD BOLT	1	STEEL	M30 x 360										
NO.	PART NAME	QTY.	MATERIAL	SPECIFICATION										

Part 8: Slotted Hex Nut

DATE:	
NO.:	
SEC.:	
TITLE:	DESIGN DRAWINGS EXAMPLE
INSTR.:	
DR. BY:	JCS
CH. BY:	
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STANDARD PART
SEE SPECIFICATION FOR PURCHASE

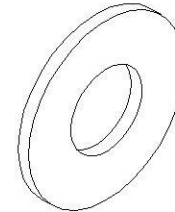
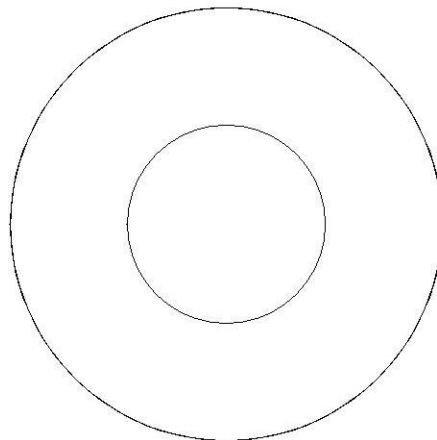


8	SLOTTED HEX NUT	5	STEEL	M30 x 3.5
NO.	PART NAME	QTY.	MATERIAL	SPECIFICATION

Part 9: Washer

DATE:	TITLE: DESIGN DRAWINGS EXAMPLE
NO.:	INSTR.:
SEC.:	DR. BY: JCS
	CH. BY:
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STANDARD PART
SEE SPECIFICATION FOR PURCHASE



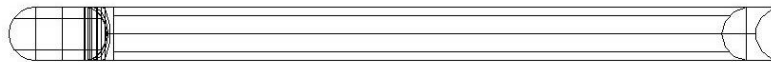
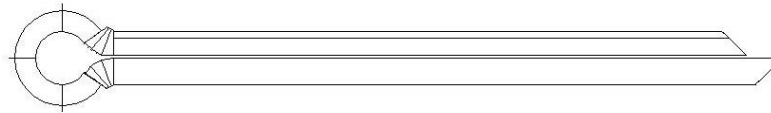
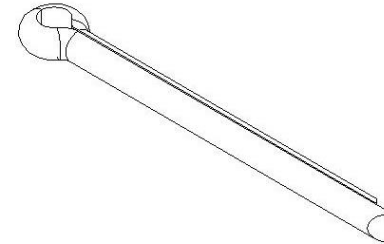
9	WASHER	10	STEEL	M30, REGULAR
NO.	PART NAME	QTY.	MATERIAL	SPECIFICATION

Part 10: Cotter Pin

DATE: _____
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 SEC.: _____
 TITLE: DESIGN DRAWINGS EXAMPLE
 INSTR.: _____
 DR. BY: JCS
 CH. BY: _____
 REV. NO.: 051994

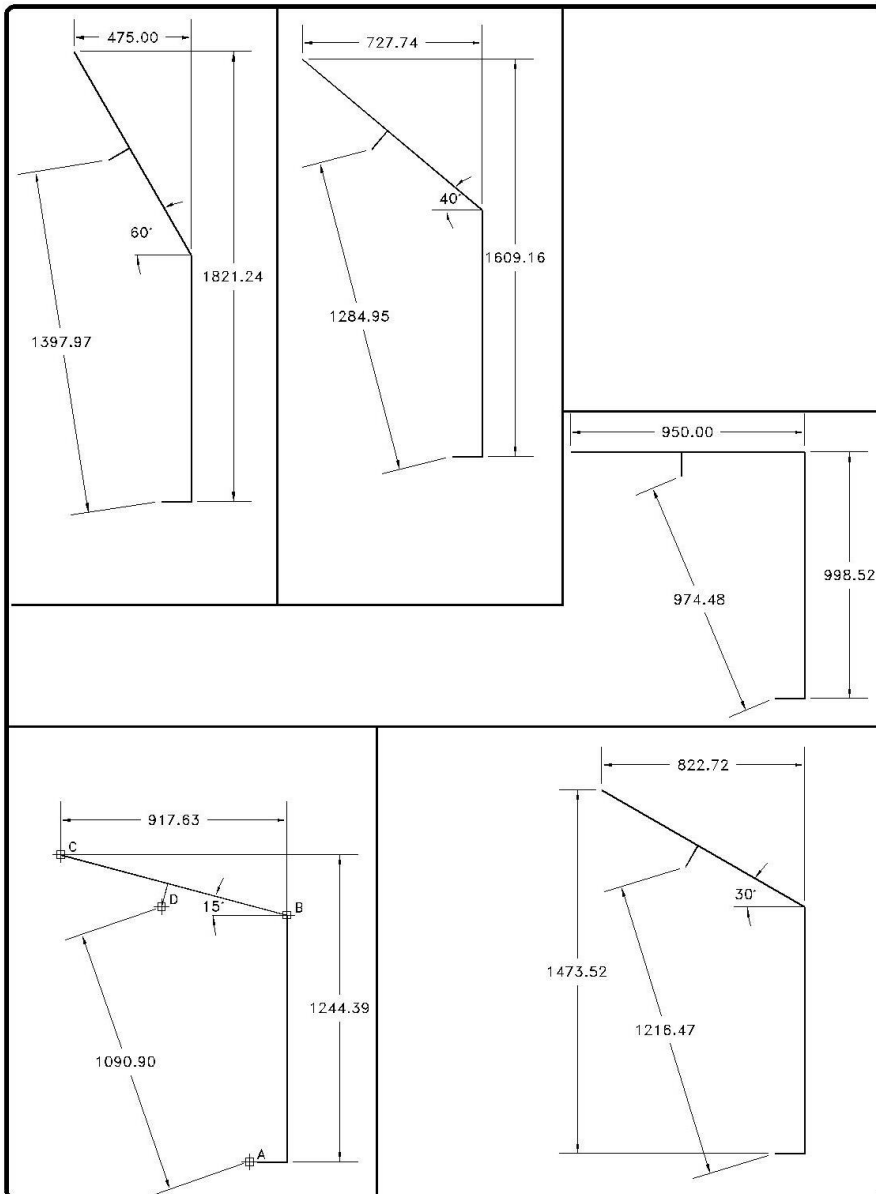
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STANDARD PART
 SEE SPECIFICATION FOR PURCHASE



10	COTTER PIN	5	STEEL	6.3mm x 45mm
NO.	PART NAME	QTY.	MATERIAL	SPECIFICATION

Layout Drawing:



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Working Drawings

Concepts: Gr 1.15a

1. TYPES OF DESIGN DRAWINGS. (REFER TO DESIGN DRAWINGS EXAMPLE)

PAGE NO.	DESIGN DRAWING TYPE(S)
2	
3	AND
13	

2. ANSWER THE FOLLOWING QUESTIONS ABOUT DESIGN DRAWINGS IN GENERAL.

A. DETAIL DRAWINGS.

- DETAIL DRAWINGS MUST INCLUDE _____, _____, AND _____.
- EXPLAIN THE PURPOSE OF DETAIL DRAWINGS.

B. ASSEMBLY DRAWINGS.

- EXPLAIN THE PURPOSE OF ASSEMBLY DRAWINGS.
- WHAT IS THE DIFFERENCE BETWEEN AN ASSEMBLY DRAWING AND A SUB-ASSEMBLY DRAWING.

C. PARTS LIST.

- EXPLAIN THE PURPOSE OF A PARTS LIST.
- A PARTS LIST INCLUDES _____ AND _____ PARTS.
- EXPLAIN THEIR DIFFERENCES.

D. OTHER DRAWINGS.

- EXPLAIN THE PURPOSE OF A LAYOUT DRAWING.
- EXPLAIN THE PURPOSE OF IDEA AND CONCEPT SKETCHES.
- COMPARE IDEA AND CONCEPT SKETCHES. INDICATE SIMILARITIES AND DIFFERENCES.

NOTES:

DATE:	TITLE: Working Drawings	PROB. GRADE
NO.:	INSTR.: jcs	1.15a
SEC.:	DR. BY:	REV. NO.: 08-0820

Concepts: Gr 1.15b

3. REFER TO THE DESIGN DRAWINGS EXAMPLE TO CALCULATE THE FOLLOWING.

A. DETERMINE THE AMOUNT OF CLEARANCE BETWEEN PARTS ① AND ②.

+ CRITICAL DIMENSION'S LIMITS OF PART ① =

+ CRITICAL DIMENSION'S LIMITS OF PART ② =

+ MIN. CLEARANCE = ②_{MIN} - ①_{MAX} = - =

+ MAX. CLEARANCE = ②_{MAX} - ①_{MIN} = - =

+ ACCORDING TO THIS DESIGN IS THERE ALWAYS GOING TO BE SOME CLEARANCE BETWEEN PARTS ① AND ②. YES OR NO

B. DETERMINE THE AMOUNT OF CLEARANCE BETWEEN PARTS ① AND ③.

CRITICAL DIMENSION OF PART ① = ±

CRITICAL DIMENSION OF PART ③ = ±

①_{MAX} = ③_{MAX} =

①_{MIN} = ③_{MIN} =

MIN. CLEARANCE = ①_{MIN} - ③_{MAX} = - =

MAX. CLEARANCE = ①_{MAX} - ③_{MIN} = - =

C. REFER TO THE DETAIL DRAWING OF PART ②. (FRONT VIEW)

NOMINAL HORIZONTAL DISTANCE BETWEEN HOLES =

NOMINAL VERTICAL DISTANCE BETWEEN HOLES =

CENTER TO CENTER DISTANCE BETWEEN HOLES.

MEASURED =

CALCULATED =

D. THE DIMENSIONS OF PART ④ CAN BE FOUND ON PAGE ____ OF THE TEXT. DETERMINE THE FOLLOWING DIMENSIONS.

NOMINAL BOLT DIAMETER =

LENGTH OF BOLT =

BODY DIAMETER LIMITS =

WIDTH ACROSS FLATS =

WIDTH ACROSS CORNERS =

E. LIST MEASUREMENTS TO CHECK THAT WILL VERIFY THE FIT OF PART ④ WITH THE REST OF THE ASSEMBLY. (DON'T CALCULATE, JUST LIST)

NOTES:

DATE:	SEC.:	TITLE: Working drawings	PROB. GRADE
NO.:	INSTR.:	jcs	1.15b
	DR. BY:		REV. NO.: 08-0820